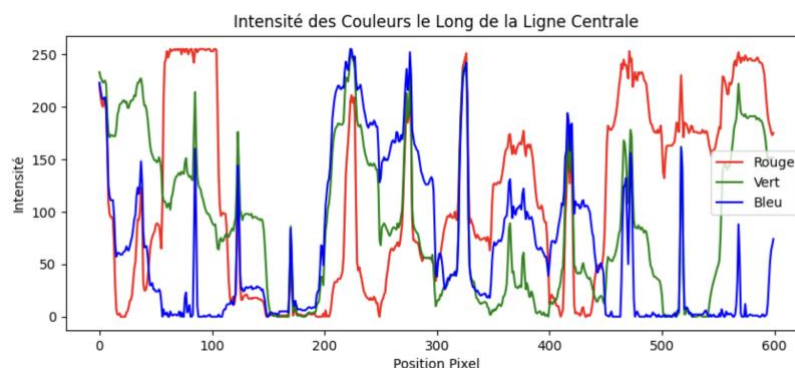


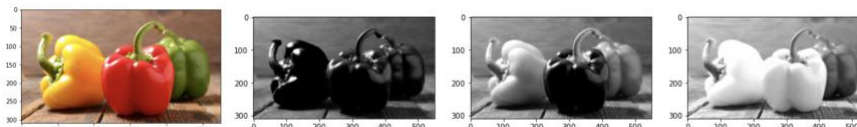
To start with colors...

Prerequisites: install Python with Notebook Jupyter and the OpenCV, Matplotlib and Numpy libraries.

- 1) Launch Jupyter and create a new Notebook. Import the OpenCV, Matplotlib (pyplot module) and Numpy libraries, and write a program to read and display a color image. Use the image Peppers or Pens, for example. Check the parameters of the `cv2.imread()` function. What are the dimensions of the image matrix loaded into memory? How is the color encoded in each pixel?
- 2) Display a horizontal band of the image, about fifty lines long and plot the intensity profiles of each RGB value along the center line on a graph below. Observe the values of the intensities corresponding to the colors displayed. Do they seem consistent with additive color synthesis?



- 3) Extract each of the 3 components R, G and B and display them as grayscale images with an appropriate title. Be careful that the tapes are read correctly by the `cv2.imread()` function, otherwise swap the tapes by the `cv2.cvtColor(image, cv2.COLOR_BGR2RGB)`.



- 4) Create another color image by swapping the stripes with the function `cv2.merge((band1, band2, band3))` and display it. You can also see the effect of duplicating the same band twice, for example (B,B,G).
- 5) Set one of the bands to zero and display the resulting color image.

6) Reverse one of the bands and recombine it with the other 2 and display the color image obtained.

7) Compare the three formulas below to convert a color image to a luminance image:

$$L = (R+G+B)/3$$

$$V = \max(R, G, B)$$

$$Y = 0,30 R + 0,59 G + 0,11 B$$

Are the results equivalent? What is the point of the last formula?

8) Apply a threshold to each RGB component and display the resulting images in binary form (black if < threshold and white if > threshold).

9) Combine the binary results obtained on each plane by the *min* and *max* operators. What do these two operators mean? Compare the results obtained with a thresholding applied to the luminance image. Display the resulting color image of the thresholding using the binary mask.

10) Color space conversion: Consult the documentation

https://docs.opencv.org/3.4/de/d25/imgproc_color_conversions.html

After converting to YCrCb, display all 3 components.

Start again with HSV.

11) Work to be done:

In your report, take care of the writing: present and justify your approach and comment on the results. Number and caption the figures, present the images in reduced resolution to limit the file size (on-screen viewing). Place the code in the appendix.

11.a) Display profiles for a line in the top color area of the MireTV.jpg image and a line in the grayscale gamut in the RGB and YCrCb spaces. How is this target designed?

11(b) Propose a method to isolate the red areas and then the green areas in the Parrots and Pens images.

11.c) Propose an algorithm to repaint the "Dedeuche" (2 CV) in green.

